GC-141-167

Smart Resume Screening & Interview Preparation Assistant

Project Scope

This project focuses on the development of a functional prototype for an AI-driven resume screening and interview preparation tool. The system supports resume ingestion, semantic analysis, candidate ranking, and interview question generation. It evaluates candidates across skills, experience, education, and projects using embedding-based similarity and LLM-generated insights.
The scope is limited to early-stage candidate evaluation and decision support. The system does not automate hiring decisions, reject candidates, or operate as a production-scale solution. It is designed for demonstration purposes within an academic environment and operates within computational and API constraints.

Abstract

The hiring process often relies on manual resume review and keyword matching, which can lead to inconsistent and biased candidate evaluations. This project introduces a Smart Resume Screening and Interview Preparation Assistant designed to improve transparency and consistency in early-stage candidate evaluation. The system allows recruiters to upload resumes and job descriptions, then uses embedding-based semantic matching and large language models to assess candidate alignment across skills, experience, education, and projects. The application generates structured rankings, explainable insights, and tailored interview questions. This project focuses on developing a functional prototype that demonstrates how AI can enhance decision support while maintaining human control in hiring workflows.

Research Questions

1. How can semantic embedding techniques improve the accuracy of resume-to-job matching compared to traditional keyword-based methods?
2. To what extent can large language models generate consistent and explainable candidate evaluations?
3. How effective is an AI-driven system in identifying candidate strengths and gaps across key evaluation categories?
4. Can automated interview question generation enhance the recruiter's ability to assess candidates more effectively?
5. How can AI be used to support hiring decisions while maintaining human control and reducing bias?

Evaluation and Validation

To address the research questions, the system was evaluated by comparing traditional keyword-based resume matching with embedding-based semantic analysis. Performance was assessed using candidate ranking accuracy, relevance of matches, and consistency of generated evaluation outputs. Results show that semantic embeddings improve contextual understanding between resumes and job descriptions, leading to more accurate candidate alignment. Additionally, LLM-generated evaluations produced consistent, structured, and explainable outputs across candidates. Each research question was addressed through system-generated outputs, including category-level scoring, overall candidate rankings, and dynamically generated interview questions aligned with identified strengths and gaps.

Comparison To Existing Solutions

Traditional Applicant Tracking Systems (ATS) rely heavily on keyword matching, which can overlook qualified candidates due to rigid filtering. While newer AI-based tools introduce automation, many lack transparency in how candidate evaluations are generated.
This system differentiates itself by combining semantic embedding techniques with LLM-generated explanations to improve both matching accuracy and interpretability. Unlike many existing solutions, the system emphasizes decision support rather than automation, ensuring human oversight remains central to the hiring process.
This approach improves consistency, fairness, and transparency in early-stage candidate evaluation.

Analysis of Process

System Architecture

- React frontend for user interaction and dashboard display
- Python (FastAPI) backend for processing and orchestration
- MongoDB for storing candidate data and outputs

Methods Used

- Resume parsing to convert PDF/DOCX into structured JSON
- Embedding generation using Sentence Transformers
- Semantic similarity scoring between resumes and job descriptions
- LLM-based evaluation using LLaMA for scoring and explanations
- Interview question generation using Gemini

Data Collection

- Sample resumes with varying formats and experience levels
- Job descriptions used as baseline comparison inputs
- Generated outputs (scores, explanations, questions) stored in MongoDB

Process Workflow

1. Upload job description
2. Upload resumes
3. Perform semantic analysis
4. Score candidates (category + overall)
5. Generate interview questions
6. Store and display results in dashboard

Results

- Successfully developed a working end-to-end prototype
- Resume parsing and semantic matching implemented across varied formats
- Embedding-based matching improved contextual relevance compared to keyword-based approaches
- Candidate ranking system produced consistent and structured scoring outputs
- LLM-generated explanations enhanced transparency by identifying strengths and gaps
- Interview questions dynamically aligned with candidate-specific attributes
- System demonstrated reliable performance across multiple test resumes and job descriptions.

Conclusion

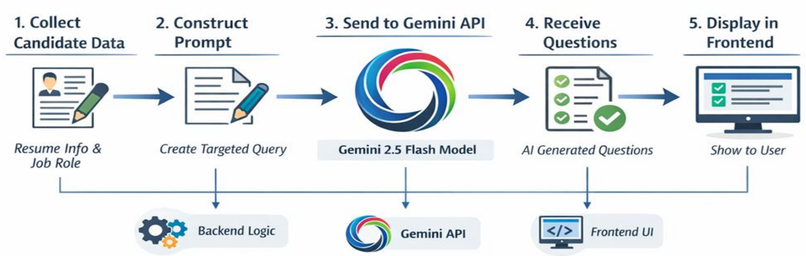
The results demonstrate that AI-driven semantic analysis improves early-stage recruitment by increasing matching accuracy, consistency, and transparency. Embedding-based approaches outperform traditional keyword methods in identifying relevant candidate qualifications. LLM-generated insights further enhance interpretability by clearly explaining candidate strengths and gaps. By combining intelligent automation with human oversight, the system provides a practical and scalable solution that addresses key limitations in existing hiring tools.

Research Outcomes

1. Semantic embeddings improved matching accuracy by capturing contextual relationships between candidate experience and job requirements, allowing relevant qualifications to be identified even without exact keyword matches.
2. The system produced structured and consistent evaluation outputs across candidates, with clear explanations that identified strengths and gaps, improving transparency in the screening process.
3. The system successfully categorized candidate qualifications across skills, experience, education, and projects, enabling more comprehensive and consistent assessments compared to manual review.
4. Generated interview questions were tailored to each candidate's background, helping target specific strengths and gaps and supporting more focused and effective interviews.
5. The system functions as a decision support tool, providing structured insights and recommendations while leaving final decisions to human recruiters, promoting transparency and reducing reliance on subjective judgment.

Resources

Project Repository: https://github.com/KSU-Capstone-Group-P12/Smart_Resume_Screening



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